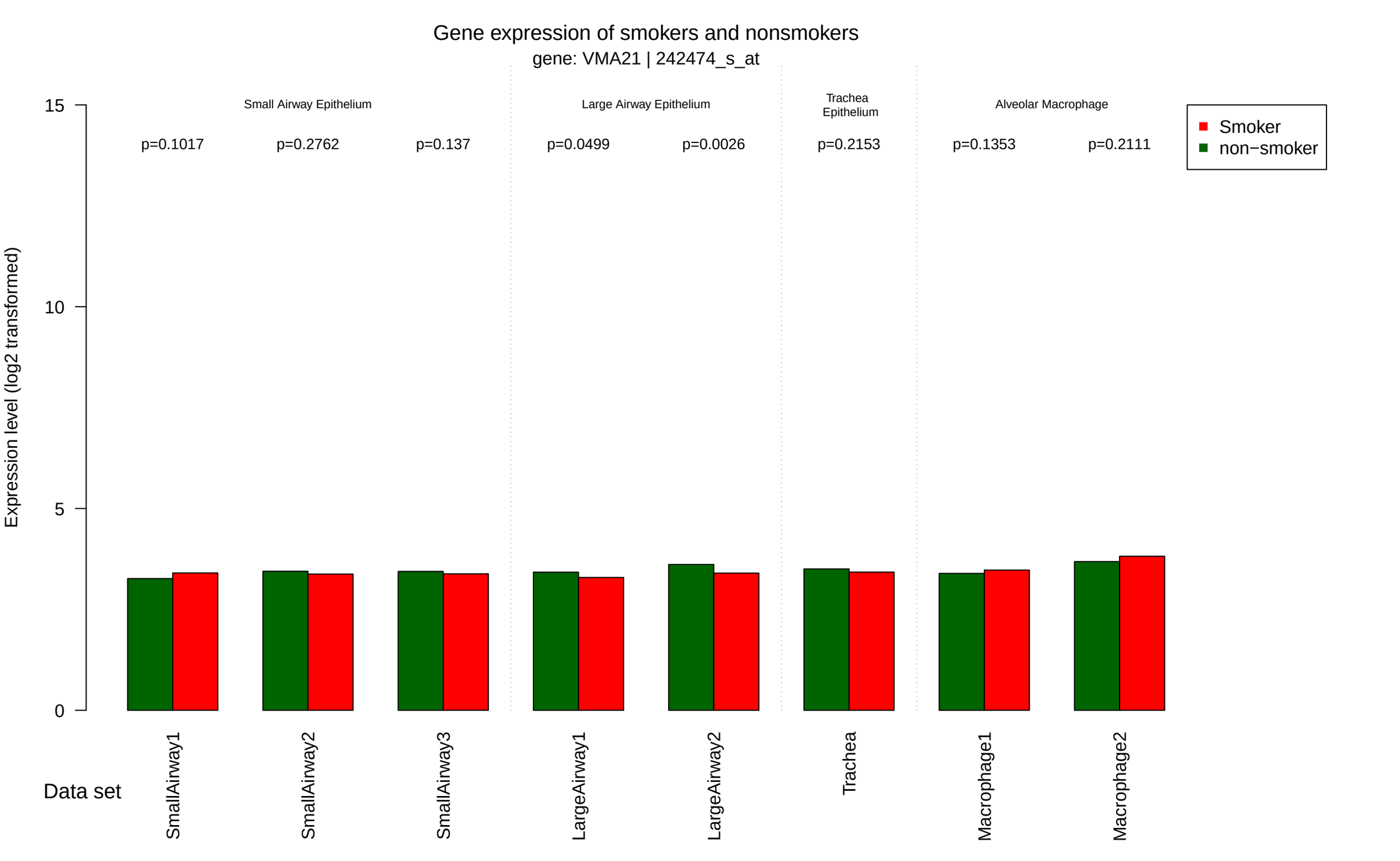
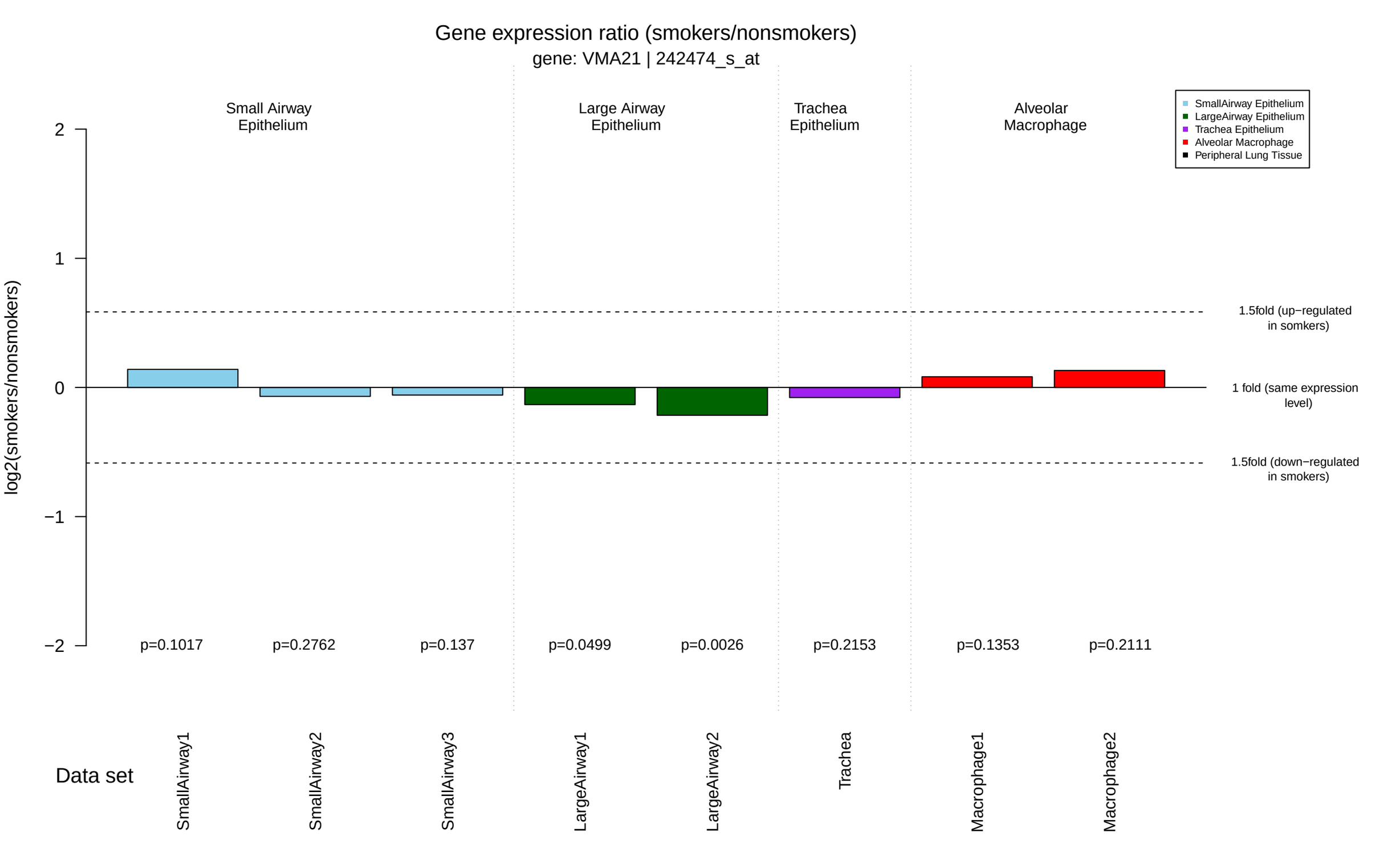
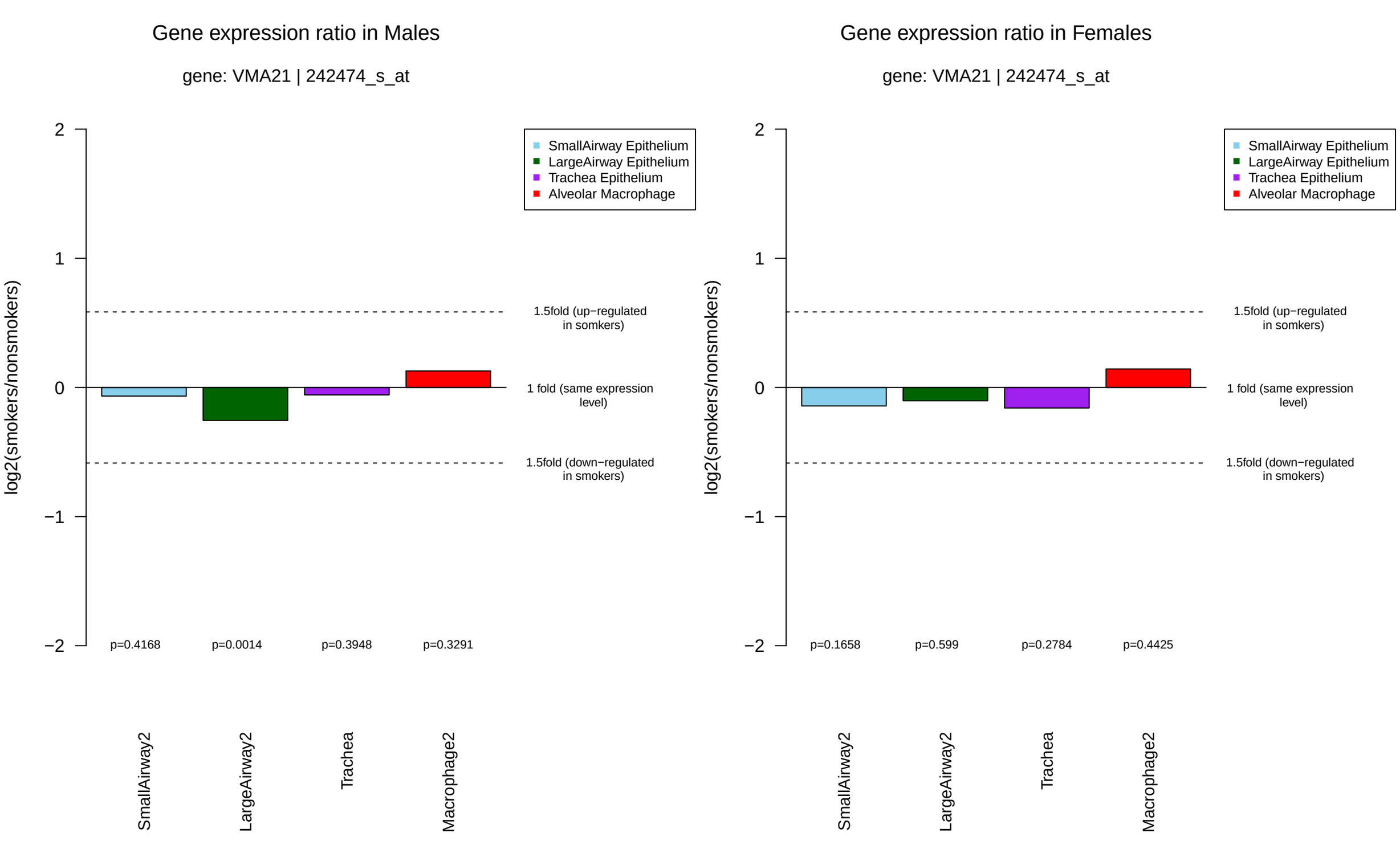
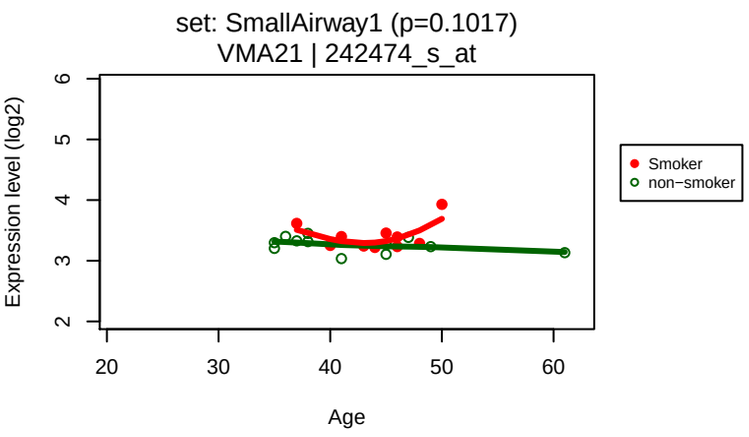
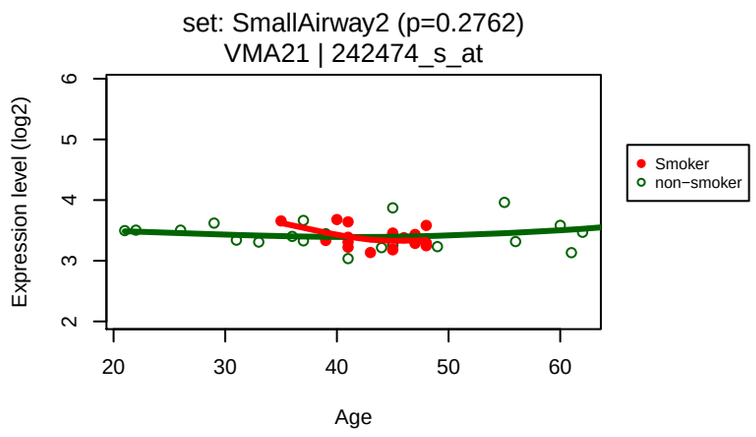
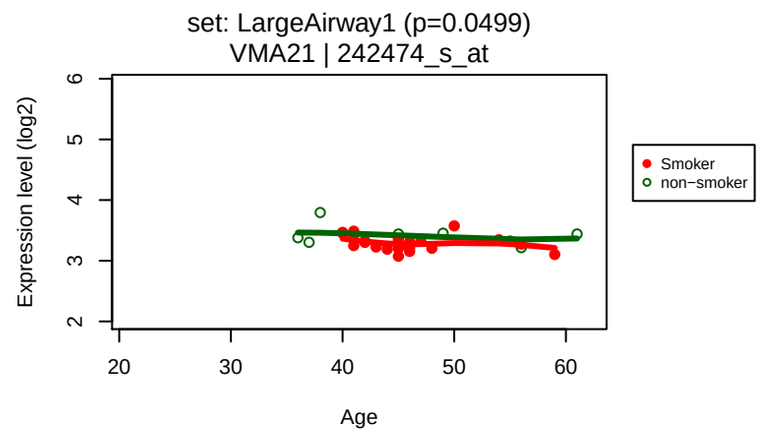

Gene expression of smokers and nonsmokers
gene: VMA21 | 242474_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1017
p=0.2762
p=0.137
p=0.0499
p=0.0026
p=0.2153
p=0.1353
p=0.2111
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

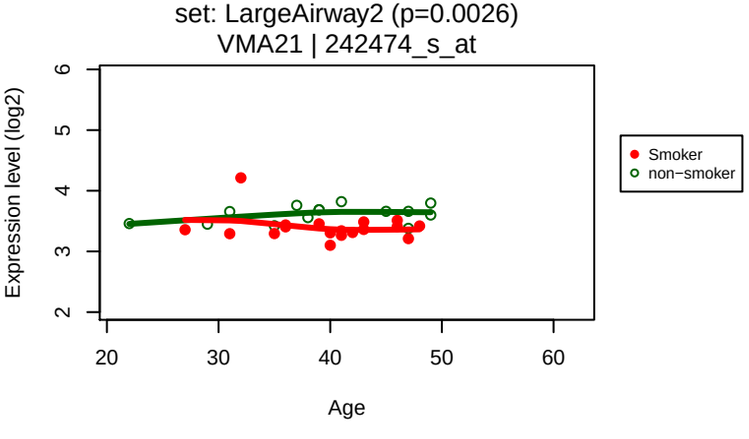

Gene expression ratio (smokers/nonsmokers)
gene: VMA21 | 242474_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1017
p=0.2762
p=0.137
p=0.0499
p=0.0026
p=0.2153
p=0.1353
p=0.2111
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

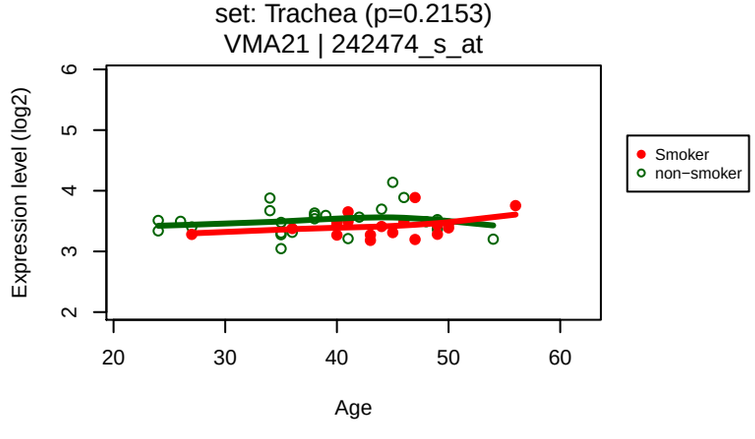

Gene expression ratio in Males
gene: VMA21 | 242474_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4168
p=0.0014
p=0.3948
p=0.3291
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VMA21 | 242474_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1658
p=0.599
p=0.2784
p=0.4425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

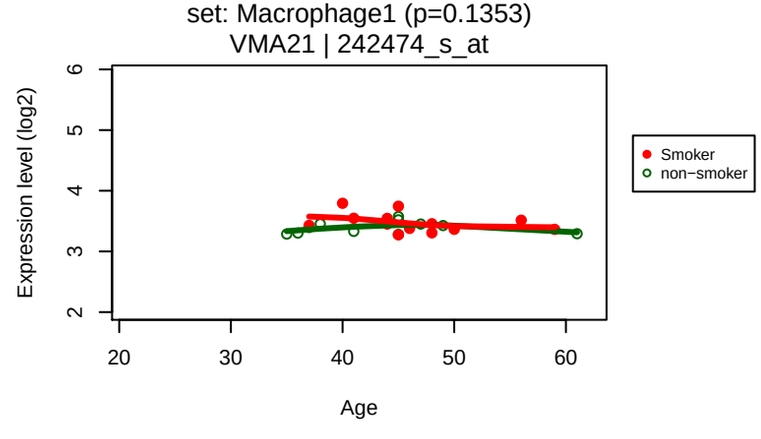

set: SmallAirway1 (p=0.1017)
VMA21 | 242474_s_at
Expression level (log2)
Age
Smoker
non−smoker

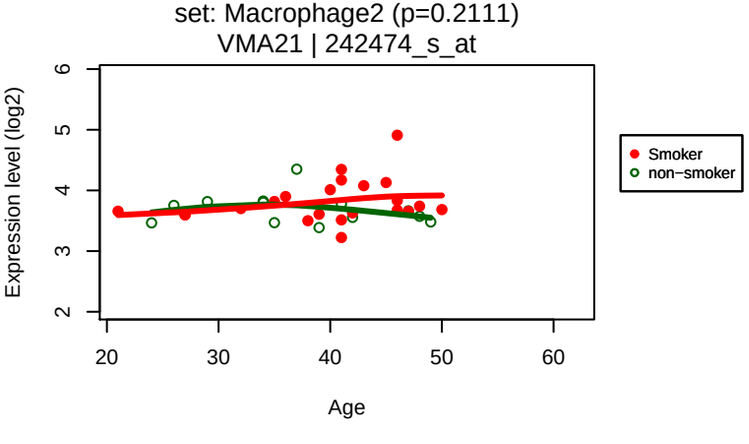

set: SmallAirway2 (p=0.2762)
VMA21 | 242474_s_at
Expression level (log2)
Age
Smoker
non−smoker

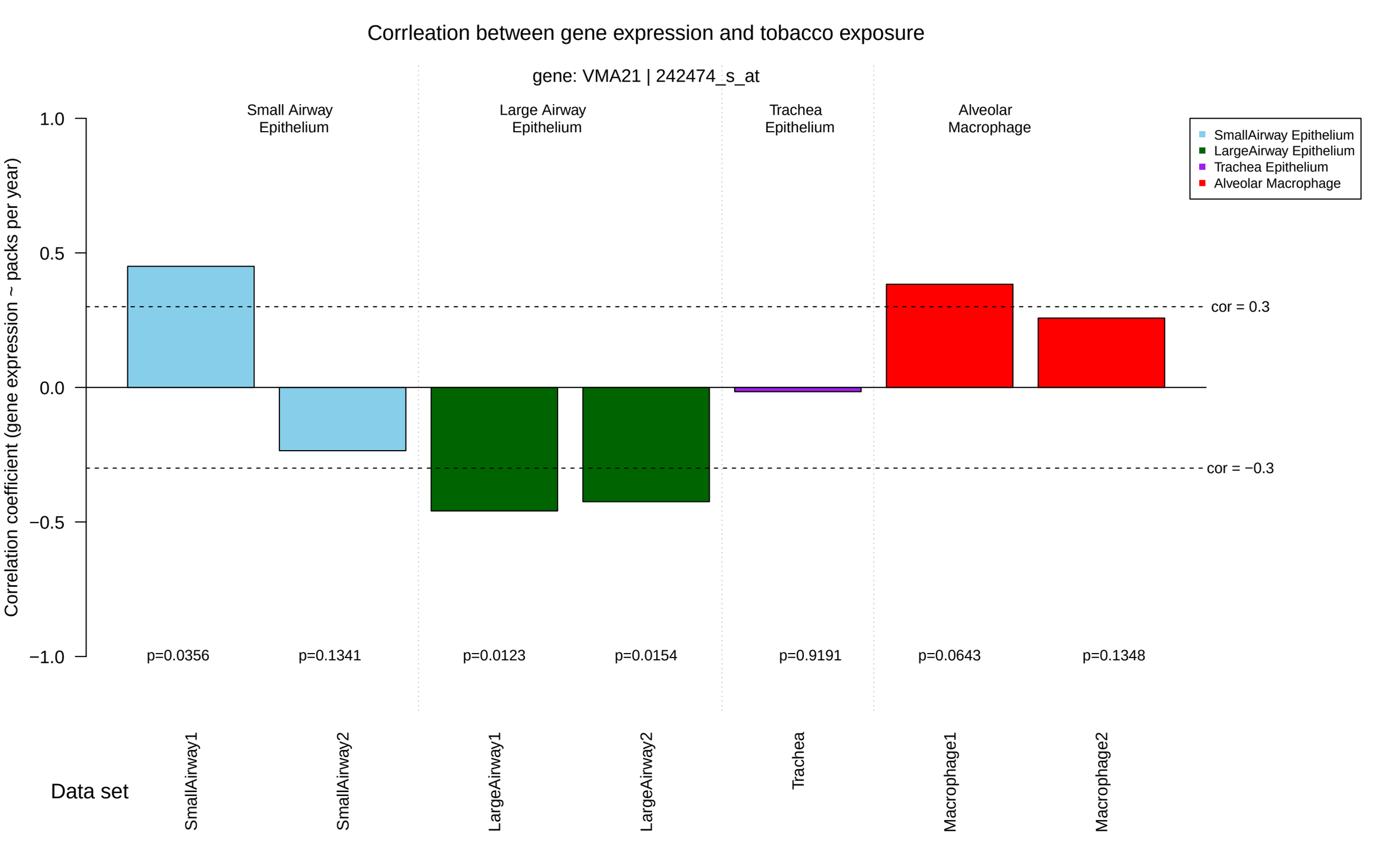

set: LargeAirway1 (p=0.0499)
VMA21 | 242474_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0026)
VMA21 | 242474_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2153)
VMA21 | 242474_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1353)
VMA21 | 242474_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2111)
VMA21 | 242474_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VMA21 | 242474_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0356
p=0.1341
p=0.0123
p=0.0154
p=0.9191
p=0.0643
p=0.1348
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2